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Educational Paper

Underpowered studies in muscle metabolism research: Determinants and considerations

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SUMMARY

Biomedical research frequently employs null hypothesis testing to determine whether an observed difference in a sample is likely to exist in the broader population. Null hypothesis testing generally assumes that differences between groups or interventions are non-existent, unless proven otherwise. Because biomedical studies with human subjects are often limited by financial and logistical resources, they tend to have low statistical power, i.e. a low probability of statistically confirming a true difference. As a result, small but potentially clinically important differences may be overseen or ignored simply due to the absence of a statistically significant difference. This absence is often misinterpreted as 'equivalence' of treatments. In this educational paper, we will use practical examples related to the effects of exercise and nutrition on muscle protein metabolism to illustrate the most important determinants of statistical power, as well as their implications for both investigators and readers of scientific articles.

Changes in muscle mass occur at a relatively slow rate, making it practically challenging to detect differences between treatment groups in a long-term setting. One way to make it 'easier' to differentiate between groups and hence increase statistical power is to have a sufficiently long study duration to allow treatment effects to become apparent. This is especially relevant when comparing treatments with relatively small expected differences such as the effect of modest changes in daily protein intake. Secondly, one could try to minimize the variance and response heterogeneity within groups, for example by using strict inclusion criteria and standardization protocols (e.g., meal provision), by using cross-over designs, or even within-subject designs where two interventions are compared simultaneously (e.g., studying an exercised limb vs a contralateral control limb) although this might limit the generalizability of the findings (e.g. such single-limb exercise training is not common in practice). In terms of data interpretation, investigators should obviously refrain from drawing strong conclusions from underpowered studies. Yet, such studies still provide valuable data for meta-analyses. Finally, because muscle protein synthesis rates are highly responsive to anabolic stimuli, acute metabolic studies are more sensitive to detect potentially clinically relevant differences in the anabolic response between treatments. Apart from further elaborating on these topics, this educational article encourages readers to more critically question null findings and scientists to more clearly discuss limitations that may have compromised statistical power.

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1. Introduction

Null hypothesis testing is the most commonly applied statistical approach in biomedical research [1,2]. In such studies, the typical null hypothesis (H_0) is that there are no mean differences between two or more conditions, whereas the alternative hypothesis (H_a) states that there is in fact a mean difference. After defining these hypotheses, an appropriate statistical test is used to compare the conditions. This test generates a *P*-value and if this value is equal to or below a predefined value denoted by α (usually 0.05 or 5%), H_0 is rejected in favor of H_a, i.e. the conditions are deemed to be (statistically) significantly different. Still, there remains a 5% chance (i.e., if $\alpha = 0.05$), or 'probability' in statistical terms, to wrongly draw the conclusion that there is a difference between conditions based on the data, whereas in fact there is no difference in the population. This scenario is referred to as a false positive result, or a type I error. Alternatively, if $P > \alpha$, H_0 is not rejected, indicating that there is no significance and as such, no evidence of a difference between the conditions.

A common misconception is that the absence of a statistically significant difference implies that the two conditions are equal. In other words: 'no evidence of a difference' is not the same as 'evidence of no difference' [3]. Again, the *P*-value simply indicates the 'probability' to get results at least as extreme as the results observed, assuming that H_0 is true. The threshold for that probability is usually set at 5%. In reality, however, one condition might be superior to the other, but the study's observations do not provide enough evidence for this. In this case the result would be referred to as false negative or a type II error. The probability of a study to statistically confirm that two conditions are *not* the same (correctly reject H_0) is defined as statistical power. The lower the probability of making a type II error (β), the larger the statistical power (1-β). In practice, the typical threshold for statistical power is 80%, corresponding to a 20% probability of making a type II error.

Importantly, statistical power is closely related to the significance criterion (α), the expected effect size (ES), and the sample size such that each parameter is a function of the other three [4]. This means that if three variables are known, the fourth can be calculated. This principle is commonly applied in sample size planning, where the required sample size is calculated for a given α and β (typically 0.05 and 0.20, respectively) and ES (typically estimated based on previous work) [5,6]. A sample size calculation is highly dependent on the study design. The underlying calculations for various study designs and corresponding examples have been described in detail previously [7–9]. In practice, researchers often use freely available software such as G*Power or Superpower to perform sample size and/or power calculations [10,11].

Surprisingly, despite the consensus that *a priori* sample size planning is a crucial component of a study design, the (correct) use and reporting thereof remains problematic in nutrition and exercise research [5,12–14]. This has resulted in a disproportionally large number of underpowered studies that are by definition less likely to significantly detect a clinically relevant difference, even when there is a difference in reality [15,16]. This is clearly problematic as the outcomes of such a study will be inconclusive and may give the impression that there is no difference between two conditions, or even be incorrectly interpreted as the two conditions being equivalent.

Here, we will discuss the most important determinants of statistical power by means of practical examples related to the effects of exercise and nutrition on muscle protein metabolism. This field of research is especially prone to underpowered studies since the sought-after effects are typically small, making them difficult to detect using the available techniques and resources. Although the latter is often out of the researcher's control, we here describe several methodological considerations to ensure a sound study design. Furthermore, emphasis will be placed on the measurement of muscle protein synthesis rates as an acute outcome measure that is highly responsive to anabolic stimuli and, therefore, sensitive enough to detect small, yet clinically relevant, differences between treatments. By means of this educational article, we aim to encourage critical thinking among readers and make scientists in the field aware of their responsibility to draw correct conclusions from their work, and not to over- or under-interpret their research findings.

2. Mind your effect size

As described, for a given α and β (i.e., power), the expected ES determines the required sample size of a study. More specifically, the larger the expected ES, the smaller the required sample size. Note that a larger ES makes it 'easier' to statistically detect a true effect, thereby increasing statistical power for a given sample size (i.e., lowering the probability of not detecting the effect when an effect exists). Thus, everything else being equal, if the mean difference between two treatments is large, it will require less participants to detect a statistically significant effect and *vice versa*. In practice, it is often financially and logistically challenging to recruit a large number of study participants. Therefore, studies should be designed such that with a relatively small number of subjects, the ES that can be statistically detected is also of clinical relevance.

The ES can be calculated in various ways depending on the study design and research question of interest [17]. For the purpose of this article, it is important to understand that the ES is not only dependent on the mean difference between treatment groups, but also the variance or standard deviation within each treatment group. We will first cover the factors that impact the mean difference, after which the factors influencing the variance will be elaborated on.

3. Study duration: effect on mean difference

The study duration often impacts the ES and, therefore, the required sample size to maintain sufficient statistical power. As an example, consider a study that investigates the effects of resistance-type exercise training (e.g. 3 sessions per week) on skeletal muscle mass. Study participants will be randomly allocated to the training group or to a no-training control group. A dual-energy X-ray absorptiometry (DXA) scan will be performed before and after the study period. What would be a suitable study duration to answer the research question? Despite the fact that long-term resistance exercise training is the most potent physiological stimulus for muscle hypertrophy [18], it takes time for muscle mass gains to accumulate, become clinically relevant, and translate to a detectable ES when compared to a (no-exercise training) control treatment (Fig. 1A) [19]. Intuitively, with a smaller stimulus (e.g. 1 as opposed to 3 training sessions per week), the ES for a given study duration will be lower, which would require many subjects to ensure ample statistical power (Fig. 1B).

As discussed, one should determine the required sample size for an experiment based on the ES that is expected from the existing literature. In longitudinal intervention studies, it is important to realize that in many cases this ES is time-dependent. For example, if one were to perform a 10-week training study, the sample size calculation should not be based on the reported ES of a prior 20-week training study, since this would result in an underestimation of the required sample size and a lack of statistical power. This may eventually lead to false negative results, i.e. concluding that the training program does not increase muscle mass, assuming there is an effect in reality.

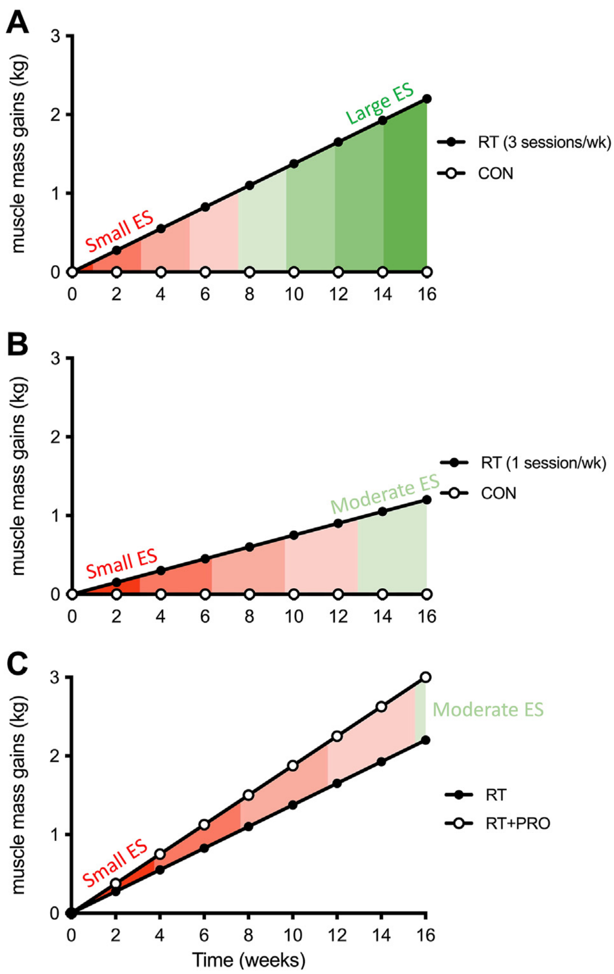

Fig. 1. Graphical depiction of muscle mass gains following prolonged resistance exercise training (RT), based on hypothetical data. (A) The effect of RT on muscle mass gains is time-dependent, with a longer study duration corresponding to a larger ES. (B) With a lesser stimulus (e.g. 1 vs 3 RT sessions/wk), the mean difference between treatments will increase more slowly over time. As such, the ES for a given study duration is smaller. (C) Similarly, when a relatively small difference is expected between treatments (e.g. RT with vs without protein supplementation), a long study duration is required before a clinically relevant and statistically detectable ES is obtained. CON, no-exercise control group; PRO, protein supplementation.

Besides exercise, dietary protein intake is considered to be the second most potent physiological stimulus to support muscle hypertrophy. Indeed, protein supplementation has been shown to further augment the adaptive response to resistance exercise training [20,21]. However, compared to the robust but heterogeneous effect of prolonged resistance exercise training, the incremental effect of protein supplementation on muscle mass gains is relatively subtle. It would therefore require a long study duration before obtaining a clinically and/or scientifically meaningful ES (Fig. 1C). Moreover, the low ES makes it difficult (i.e., requires a very large sample size) to statistically confirm the added benefit of protein supplementation on muscle mass accretion on top of the exercise training-induced improvements.

Note that finding a statistically significant effect should not be the primary objective in scientific research. In essence, even the smallest ES can become statistically significant if the sample size (and hence statistical power) is large enough. Therefore, in the process of designing a study, it should first be considered what ES would be considered clinically and/or scientifically relevant. Next, based on previous research in the field, one might be able to estimate the required study duration to obtain that effect. Finally, a sample size calculation should be performed so that the study will have sufficient statistical power to test the hypothesis. If, from the sample size calculation, it becomes apparent that the required number of subjects is not feasible (i.e. exceeds the financial/logistical resources), one might consider a more prolonged study duration to increase the expected ES and thereby limit the required sample size. However, this is only a valid alternative if the predefined research question does not entail a time-component. It should also be noted that a longer study duration will increase subjects' burden, potentially making it more challenging to recruit subjects and to ensure proper compliance to the intervention.

Importantly, there are many more factors that impact the ES and (indirectly) statistical power. Therefore, one cannot determine whether a study was appropriately powered solely based on the duration of that study. In the following paragraphs we will outline several other important factors that modulate the ES and discuss how they are interrelated.

4. Diminishing returns: effect on mean difference

The effect of protein supplementation during prolonged resistance exercise training on skeletal muscle mass accretion is dose-dependent, which impacts the expected ES and therefore has implications for sample size calculations. To illustrate this, consider a study in which participants will be randomly allocated to daily supplementation with a higher or a lower dose of protein during a 12-week resistance exercise training program. DXA scans will be performed before and after the intervention to assess lean body mass gains. Previous work suggests a dose–response relationship between protein intake and muscle mass gains during resistance exercise training with diminishing returns at higher doses [22,23]. Based on these findings, comparing the supplementation of 10 vs 20 g of protein would most likely result in a larger ES than when comparing higher doses of protein (e.g. 40 vs 50 g of daily protein supplementation), despite absolute differences (10 g) being identical (Fig. 2). The latter comparison would therefore require a larger sample size to ensure sufficient statistical power. Similarly, the ES

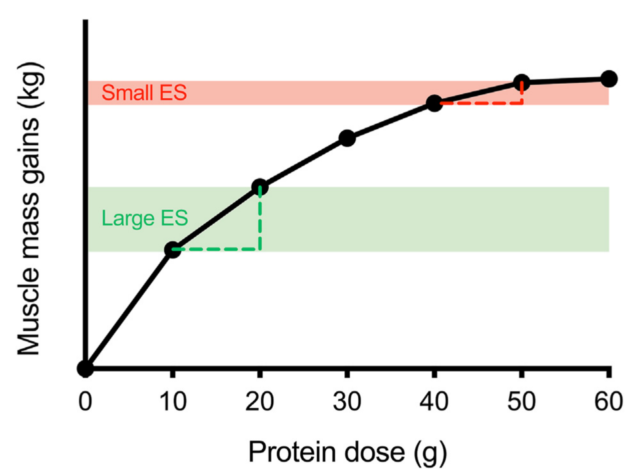

Fig. 2. Graphical depiction of the concept of diminishing returns based on hypothetical data. The added benefit of additional protein during prolonged resistance exercise training plateaus at higher doses. Therefore, an exercise training study comparing the impact of supplementing 40 vs 50 g of protein (red area) corresponds to a smaller effect size (ES) as compared to the impact of 10 vs 20 g of protein (green area), despite the same absolute differences in protein supplementation. As a result, for a given sample size, the former study is less likely to detect a statistically significant difference between treatments than the latter study.

of protein supplementation will likely be lower when habitual protein intake is already high [21].

The example of protein supplementation during resistance exercise training is further complicated by other determinants such as protein quality. Plant-based proteins are typically of a lower quality when compared to animal-based proteins and have been shown to have lesser anabolic properties [24]. Accordingly, the expected ES of supplementing 20 g of plant-derived protein per day during resistance exercise training on muscle mass gains is lower when compared to supplementing 20 g of animal-derived protein. Moreover, the isolated effects of protein quality will by definition be smaller than the already-small ES of protein supplementation *per se*. A study on the impact of supplementing two different types of protein (e.g., 20 g soy vs 20 g milk protein supplementation) on muscle mass gains during prolonged resistance exercise training would require a long study duration before obtaining a clinically relevant ES. Accordingly, a sufficiently large sample size is required to ensure ample power to statistically confirm that effect.

5. Within-subject design: effect on variance

So far, we have considered the ES as being an indicator of the mean difference between two groups. However, apart from the mean difference, the ES also depends on the variance within each treatment group, i.e. how much the individual data points differ from the treatment group mean. For a given difference between two group means, the smaller the variance (or standard deviation) within each group, the larger the ES and *vice versa* [4] (Fig. 3). Therefore, efforts to minimize variance will be reflected in obtaining a larger ES.

The variance depends on a multitude of both intrinsic and extrinsic factors, several of which are modifiable to some extent [25,26]. Extrinsic factors such as diet and physical activity should be taken into consideration especially when these factors have an acute effect on the outcome measure of interest. For example, study participants can be instructed to arrive in a fasted state or even be provided with a standardized meal or diet to consume prior to key measurements. Furthermore, participants may be asked to refrain from vigorous physical activity for a given amount of time and come to the laboratory by car or public transport. By providing all participants with identical pre-testing instructions, some variability between participants introduced by external factors may be reduced. Ultimately, which criteria apply strongly depends on the aim and design of the study, since in some cases it is more appropriate not to disturb the subjects' habitual lifestyle. Some further guidance on which factors to standardize can be found elsewhere [27].

In contrast to extrinsic factors, intrinsic factors such as sex, age, and genetics are non-modifiable. Still, their potential influence on the outcome parameters should be acknowledged when designing a study [26]. By using strict inclusion criteria, a more homogenous study population can be obtained, resulting in less inter-individual variability and, as such, greater statistical power for a given sample size and mean difference between treatment groups. However, a limitation of this approach is that the generalizability of the findings may be lowered, since the outcomes in a very specific study population may not be representative of the general population. Depending on the study aims, a broader and thus more generalizable population may be investigated. In such cases, potentially greater data variance may be compensated for by increasing the sample size to allow sufficient statistical power.

Another approach that is commonly applied in lifestyle intervention studies is the cross-over design, where subjects complete a sequence of two or more interventions separated by a wash-out period [28]. By doing so, inter-individual (between-subject) variability is omitted, and only intra-individual (within-subject, day-to-day) variability remains [29,30]. This increases statistical power and as a result, cross-over studies generally require less participants than parallel group studies to demonstrate a given effect [28]. When using a cross-over design, it is crucial that the wash-out period is sufficiently long, so that no carry-over occurs from one intervention to the other [30,31]. Moreover, external factors should be matched so that the experimental conditions are kept identical between repeated measurements. For instance, Malowany et al. [32] assessed protein requirements in resistance-trained females in a cross-over design, with participants completing a total of 6–7 metabolic trials. Before each trial, participants underwent a two-day standardization period with complete diet provision. To further reduce day-to-day variation, all trials were performed in the same phase of the menstrual cycle and participants reported to the laboratory in an overnight fasted state. When employed correctly and in the appropriate setting, the cross-over design offers a powerful alternative to the traditional parallel group design.

Finally, the unilateral model, where (within each subject) both limbs are randomly assigned to different conditions or interventions, removes both intrinsic and extrinsic variation and leads to greater statistical power than parallel group and cross-over designs (Fig. 4). This model has effectively been applied to study the effects of exercise, nutrition, and immobilization on skeletal muscle metabolism [33]. Depending on the context, the external validity of

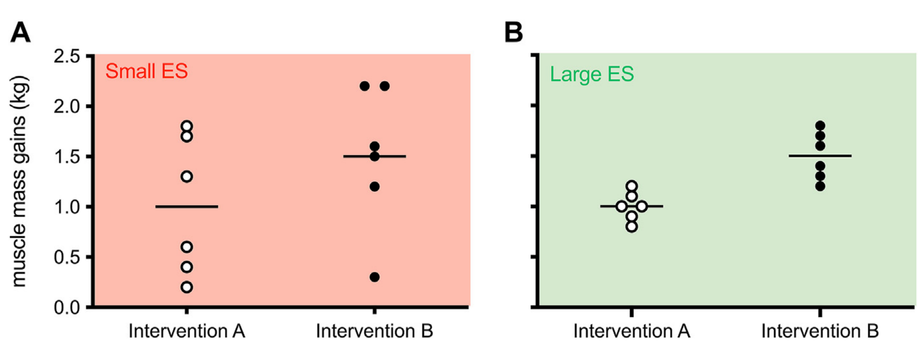

Fig. 3. Graphical illustration of muscle mass gains following two different interventions based on hypothetical data. Horizontal lines indicate the group mean and dots depict individual observations. (A) Although on average, intervention B appears to produce greater muscle mass gains compared to intervention A, the large variation of the observations within each group leads to a small effect size (ES). (B) Due to the smaller variation in both intervention groups, the exact same mean difference between interventions A and B will lead to a larger ES (standardized mean difference), making it 'easier' (i.e., detectable with a smaller sample size) to detect a statistically significant difference. Note that the same mean differences result in a different ES in these two examples. Hence, the mean difference should also be interpreted in terms of clinical relevance and not merely statistical significance.

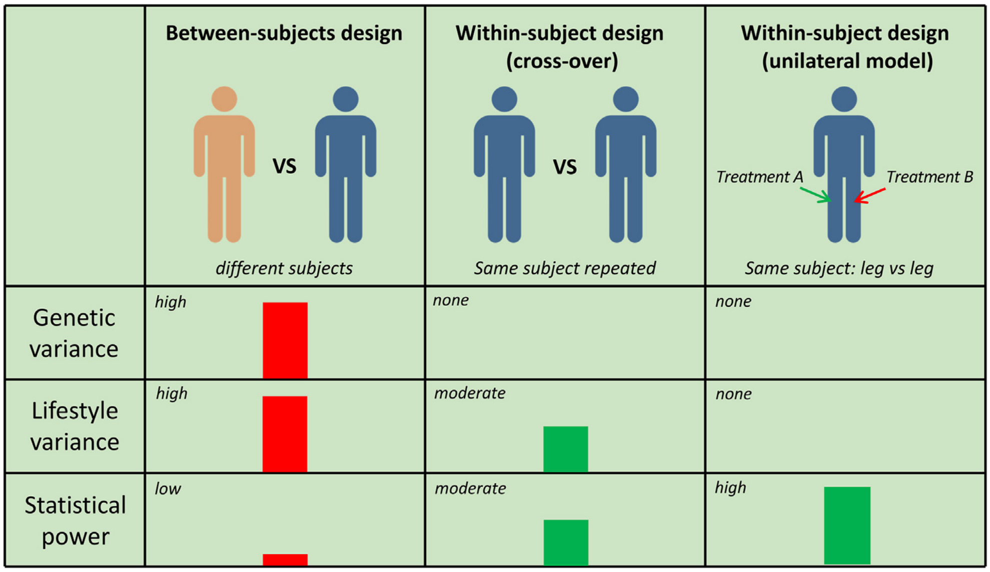

Fig. 4. Schematic overview of different study designs and how they affect data variance and, therefore, impact statistical power.

such studies might be lower. Further, any potential carryover to the contralateral limb should be acknowledged, depending on the outcome of interest [34]. However, the unilateral model is an excellent time- and cost-effective design in proof-of-principle studies. We strongly encourage scientists in the area of muscle protein metabolism (and beyond) to thoroughly consider the benefits and limitations of each of the above study designs when setting up new studies and encourage readers to take these considerations into account when interpreting published research outcomes.

6. Measurement techniques: effect on variance

Apart from the study design, the applied methodology also plays an important role in data variance. In this respect, poor reliability (i.e., large test-retest variability) of the measurement instrument can cause random variation in the data that cannot be distinguished from biological variation. To assess skeletal muscle mass, a wide variety of methods can be applied, the most accurate and reliable of which are typically also the most expensive and the least widely available. Based on this trade-off, DXA is currently the most widely applied technique to assess lean body mass, as a proxy for skeletal muscle mass [35]. However, when compared to computed tomography (CT) and magnetic resonance imaging (MRI), DXA is less capable of detecting (small) between-group differences [36] as well as pre-to-post changes in muscle mass following prolonged interventions [37]. Thus, when using DXA it will likely be more difficult to detect a significant intervention-effect, especially when the (expected) magnitude of the effect is rather low.

Similar to the sample size, study duration, and study design, the sensitivity of the measurement instrument is just another contributor to statistical power. Shortcomings in any of these variables may be compensated for by altering another variable to optimize statistical power. Therefore, it is not always necessary to use the most sensitive outcome measure as long as the ES is large enough to be detected using the preferred approach.

Techniques such as DXA, MRI, and CT can be used to assess changes in skeletal muscle mass (or a proxy thereof) over a more prolonged period of intervention. Such studies provide valuable insights in the long-term efficacy and feasibility of interventions but are typically difficult to coordinate. A long study duration increases the burden of the subjects, which often compromises adherence and compliance to the study procedures or may even prevent people from wanting to participate at all. Besides the practical complexity, long-term intervention studies often lack the statistical power required to draw firm conclusions regarding the efficacy of nutritional and exercise interventions, especially when relying on less sensitive outcome measure such as (whole-body) lean mass.

For instance, in a meta-analysis conducted by Cermak et al. [20], it was demonstrated that protein supplementation further augments the increases in lean body mass following prolonged resistance exercise training. However, from the 22 individual RCTs included, only 4 studies reported a statistically significant benefit of protein supplementation on the exercise training-induced gains in lean body mass (Fig. 5). While the mean effect of 0.7 kg lean body mass gains can be considered clinically meaningful, most individual studies (18/22) were unable to demonstrate a statistically significant effect. Still, the vast majority of these studies (15/18) observed a mean effect that was in favor of protein supplementation versus placebo. This could be an indication that some or all of these studies were underpowered, although the heterogeneity in study designs may also explain part of the discrepancy between studies. For example, an individual study with a relatively weak stimulus (e.g., low protein dose, short intervention duration, etc.) is simply less likely to induce a (significant) effect than a study with a stronger stimulus. Even though a statistically significant effect in such a 'weak stimulus study' may be detected when ample subjects were recruited, the clinical relevance of such an effect may still be questioned.

From this meta-analysis example, it is clear that underpowered studies still add valuable data to the literature. In a meta-analysis, relevant articles are systematically selected, after which data from each individual study is extracted, converted to the same metric (if possible), and given a certain 'weight'. This weighing ensures that studies with a greater sample size and/or smaller variation contribute more to the pooled mean difference. To facilitate utilization of previously collected data in meta-analyses, researchers should always report their data in a complete manner, that is reporting the means, standard deviations and sample sizes for each condition and time point. Alternative metrics such as medians, standard errors, confidence intervals, and effect sizes can

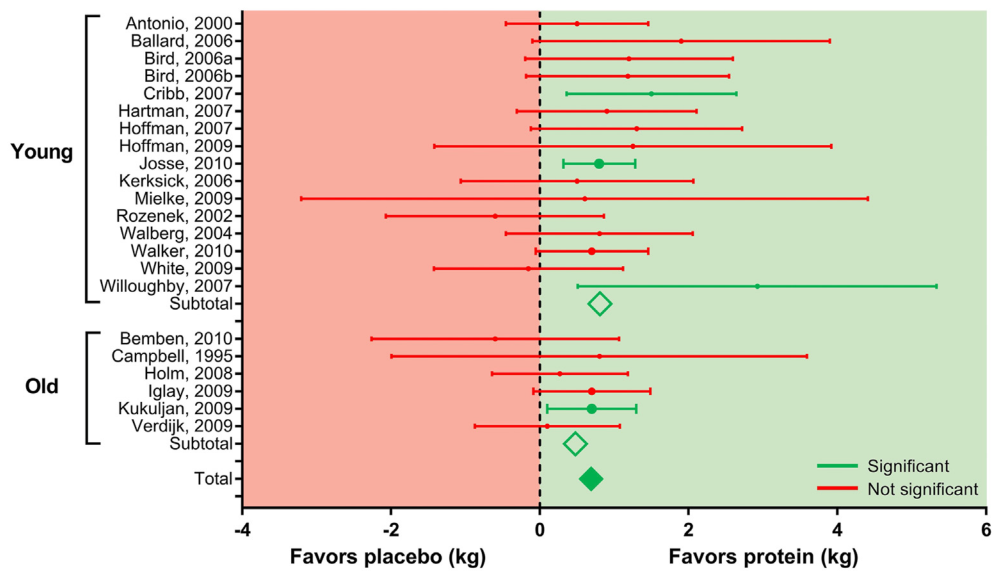

Fig. 5. Forest plot adapted with permission from Cermak et al. [20] on the effect of protein supplementation during prolonged resistance exercise training on lean body mass gains in young and older adults. Dots and horizontal lines depict the point estimate and 95% confidence interval, respectively, of each individual study outlined on the left. The size of each dot is proportional to the relative weight of the individual study. Filled and unfilled diamonds represent the total and subtotal weighted mean differences, respectively. Although the pooled data demonstrate that protein supplementation further augments lean body mass gains (depicted in green), 18 out of 22 individual studies (depicted in red) were unable to report a statistically significant benefit of protein supplementation on lean body mass gains during prolonged resistance exercise training.

also be used for meta-analyses as long as it is clearly indicated how the data are expressed [17,38]. Data sharing (either open source or upon request) is another way of maximizing the utility of previously collected data for meta-analyses.

7. Muscle protein synthesis

As discussed earlier it is challenging (i.e., requires a large sample size) to statistically detect the added effect of protein supplementation on resistance exercise training-induced gains in lean body

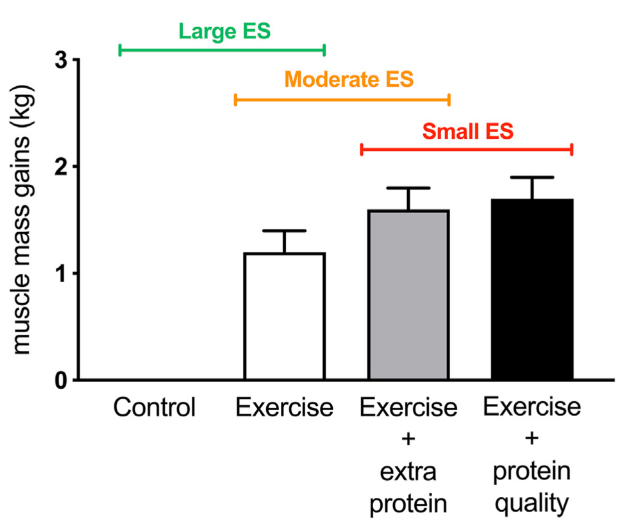

Fig. 6. Graphical illustration of different long-term interventions and their hypothetical impact on muscle mass gains. When the expected ES for a comparison is small, it is more difficult to detect a statistically significant difference. Hence, it should first be determined whether the expected difference would be clinically and/or scientifically relevant. If so, the study should be designed to compensate for the small expected ES (large number of subjects, long duration, sensitive outcome measure, etc.) to ensure sufficient statistical power.

mass. This issue will become even more evident when comparing the impact of different protein sources [39], timing strategies, or distribution patterns [40,41] of protein supplementation during prolonged exercise training where smaller ESs are expected and any potential impact would be even more challenging to detect (Fig. 6).

An alternative to measuring small changes in muscle mass throughout a more prolonged exercise training intervention is to measure the acute muscle protein synthetic response to the intervention(s) of interest. The process of muscle protein synthesis (MPS) reflects the adaptive response of skeletal muscle tissue to anabolic stimuli, thereby allowing muscle conditioning. In support, MPS rates have been shown to be highly responsive to anabolic stimuli such as exercise [42,43] and protein intake [44–46]. This responsiveness translates to a large ES, which makes that acute metabolic studies are typically more sensitive than longer-term studies in detecting the effects of exercise and dietary protein ingestion on skeletal muscle reconditioning (Fig. 7). The greater sensitivity of acute metabolic studies can in part be explained by the fact that such studies are typically performed under highly controlled and standardized experimental conditions, whereas long-term measurements are prone to variation due to differences in daily-life conditions between subjects. Besides consistently showing that protein ingestion augments MPS rates following resistance exercise [22,46,47], acute MPS studies have effectively been applied to define the impact of protein dose [22], protein type [48,49], and protein intake timing strategies [50] as well as different exercise training modalities [51] on MPS rates.

In terms of interpretation, if intervention A leads to a greater increase in MPS rates compared to intervention B, intervention A may be more effective in supporting muscle conditioning upon more prolonged exposure to the stimulus evaluated. Protein ingestion has been shown to increase MPS rates during recovery from resistance exercise, which over time may translate to enhanced muscle hypertrophy depending on the context and the duration of the intervention. Importantly, acute MPS studies should be interpreted qualitatively rather than quantitatively, i.e. 20%

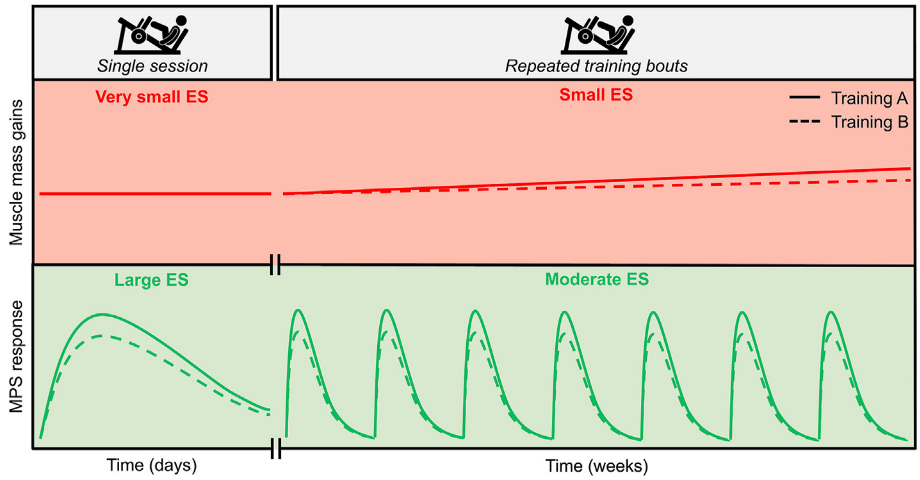

Fig. 7. Graph illustrating hypothetical changes in muscle mass (red area) and muscle protein synthesis rates (green area) following a single resistance exercise session as well as in response to resistance exercise training, consisting of multiple, successive resistance exercise bouts. Resistance exercise training A (solid line) and Resistance exercise training B (dashed line) represent the two treatments to be compared. The mean difference in muscle mass gains between the two treatments after several weeks of resistance exercise training is low, hence the small ES. However, there is a relatively large mean difference in the acute MPS response between the two treatments (hence the large ES), which may translate to greater hypertrophy following more chronic exposure. Measuring the MPS response following repeated resistance exercise bouts over a period of weeks would result in a moderate (i.e., intermediate) ES, since after each individual exercise bout, MPS rates will return to baseline in both conditions. It would take several months before the cumulative acute differences would translate into meaningful differences in muscle mass gains between treatments. Therefore, when small but clinically relevant differences between treatments are expected, acute metabolic studies provide greater sensitivity when compared to long-term studies. ES, effect size; MPS, muscle protein synthesis.

higher MPS rates do not necessarily translate to 20% greater muscle mass gains [52]. Rather, MPS is reflective of muscle protein conditioning, and the phenotypic result largely depends on the specific sets of proteins that are being expressed and the balance between the synthesis and breakdown of these proteins. Whereas MPS is largely increased during recovery from both resistance as well as endurance type exercise, the phenotypic response is quite different as can be appreciated when comparing the physique of a bodybuilder with that of a marathon runner.

Methodologically, the measurement of MPS rates requires the intravenous or oral administration of a stable isotope tracer as well as the collection of blood samples and muscle biopsies. For a complete overview of the principles and applications of stable isotope research, the reader is referred to a number of recent reviews on the topic [52–54]. Given the invasive nature and high costs of stable isotope tracer experiments and subsequent tissue analyses, its application is not as widespread as compared to commonly used imaging techniques such as DXA. However, the high degree of sensitivity and mechanistic insight that acute tracer experiments provide make this the preferred approach when subtle, physiologically relevant effects are being investigated. Ultimately, it is the combination of both acute and long-term intervention studies that move the field forward and make up a solid body of evidence combining mechanistic understanding with clinical endpoint validation.

8. Less is more

Recall that statistical power depends not only on the sample size and the ES but also on the significance criterion or type I error (α), which is typically set at 0.05. When two treatments are compared, this means that there is a 5% chance of a false positive result, i.e.

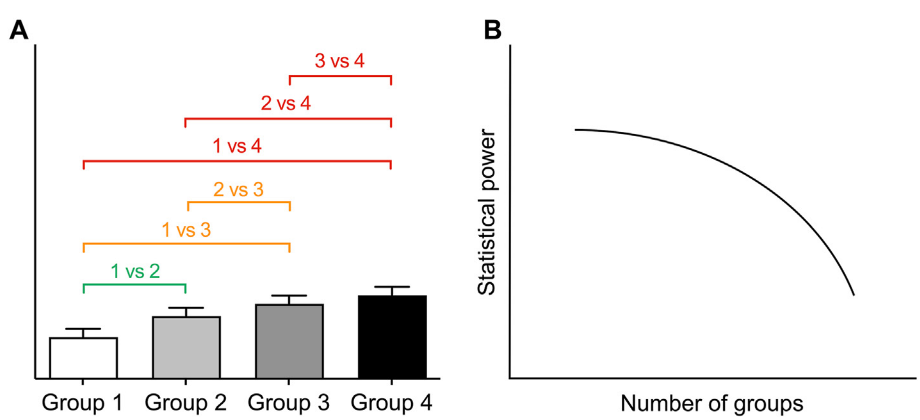

Fig. 8. Graphical illustration of the relation between the number of (treatment) groups, the number of comparisons, and the statistical power for a given sample size based on hypothetical data. (A) Increasing the number of (treatment) groups from two to three adds two additional comparisons (yellow) whereas a fourth group adds even three more comparisons (red). (B) With each additional (treatment) group, statistical power decreases exponentially unless compensated for by increasing the sample size.

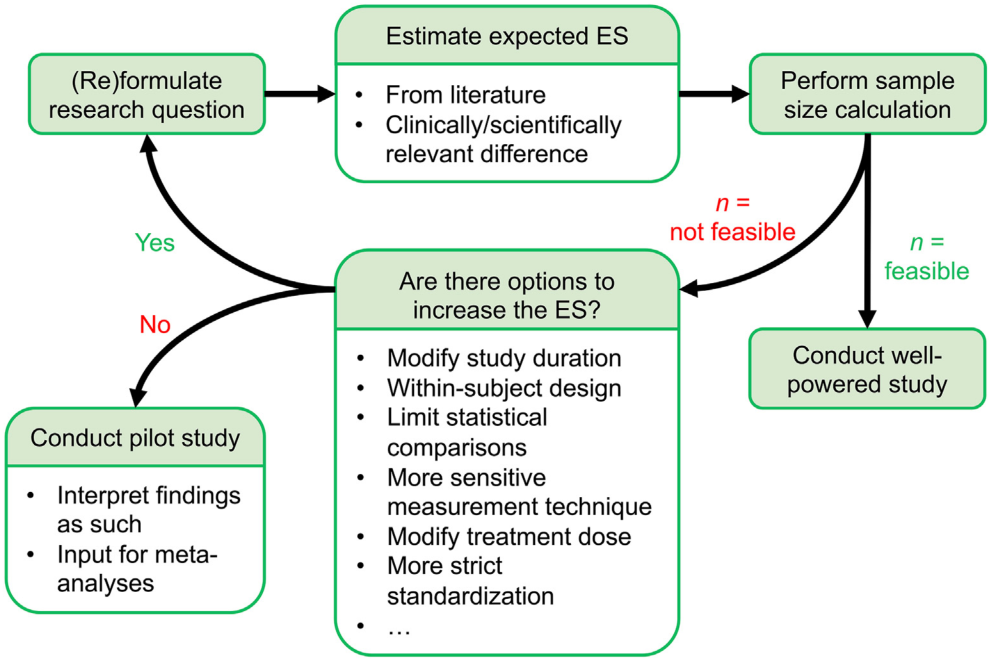

Fig. 9. Flow chart on how to design studies with adequate statistical power, including consideration of factors that modify the expected effect size (ES).

finding a difference even though there is no real difference. This error rate is generally considered acceptable. However, when three treatments are compared, each of the three comparisons (A vs B, B vs C, A vs C) runs a 5% ‘risk’ of making a type I error. In such instances, one should correct for multiple comparisons, for example by means of a Bonferroni correction [55]. In this case, one would consider each of the three comparisons significantly different only if the *P*-value is below $0.05/3 = 0.017$. Now that the probability of a false positive finding decreases, the probability of a false negative finding increases, i.e. statistical power is strongly reduced.

In the process of designing a study, it may be tempting to include more than two treatment groups assuming that it will provide more information than just a single comparison. However, since each additional treatment or group decreases statistical power exponentially, including more than two groups might actually provide less information (i.e. inconclusive results) due to a lack of statistical power (Fig. 8). Thus, unless compensated for by strongly increasing the sample size or the ES (longer duration, more sensitive measurement techniques, etc.), it is in many cases advisable to limit an experiment to a single comparison when a relatively small ES is expected. Ultimately, the number of comparisons depends on the research question of interest, but the statistical consequences of multiple comparisons need to be acknowledged, i.e. the sample size needs to be adjusted accordingly.

In some instances it may be justifiable not to correct for multiple comparisons, for example in case two treatment arms (A and B) are only compared to a control arm (C) and not to each other, i.e. A vs C and B vs C, but not A vs B, although this remains a matter of debate. Some argue that correcting for multiple comparisons is necessary for all types of research, whereas others believe it only applies to confirmatory, but not exploratory studies [56]. In other words, exploratory studies could apply a more lenient statistical approach to be able to identify interventions that are promising but require confirmation in future studies. Using either approach, adequate pre-registration of the statistical plan is crucial to ensure an unbiased statistical evaluation of the data.

9. Conclusions

Since changes in muscle mass occur at a relatively slow rate, it can be practically challenging to have sufficient statistical power to detect potential differences between treatments. As such, investigators should perform and report power calculations and design studies with consideration of modulating factors such as the sample size, study duration, treatment dose, measurement sensitivity, standardization, number of comparisons, study design (between vs within-subject), etc (Fig. 9). Reporting ES estimates rather than solely *P*-values also aids in proper data interpretation. When it is not feasible to conduct an adequately powered study, the work can still provide valuable data for inclusion into meta-analyses. However, it is important to indicate that the study was underpowered and/or refer to it as a pilot study to prevent the reader from incorrectly concluding that the treatments were equivalent. To ensure correct interpretation of null-findings, readers should assess whether an appropriate sample size calculation was performed and take note of all methodological choices (Fig. 9) that may have impacted the study's power. Metabolic fluxes (e.g. MPS rates) often are more responsive to anabolic stimuli than clinical endpoints (e.g. muscle mass). Therefore, acute mechanistic studies are more powerful in detecting small, yet clinically relevant differences in anabolic potential between treatments and may reveal interventions relevant for further study in a longer-term setting. Finally, although the examples used herein center around the field of muscle protein metabolism, the same principles apply to other fields. A basic understanding of the outlined statistical principles is essential both in the process of properly designing research studies as well as interpreting their research outcome.

Author contribution

Conceptualization: DCJH, MWB, BvH, SV, LBV, LJCvL, JT. Visualization: DCJH, MWB, JT. Writing – original draft: DCJH, JT. Writing – review & editing: MWB, BvH, SV, LBV, LJCvL.

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Declaration of competing interest

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